

11	C	From kinetic theory of gas, $p = \frac{1}{3} \rho \langle c^2 \rangle$ Since the oxygen gas is 16 times as dense as hydrogen gas, the corresponding root-mean-square speed of its molecules must be 4 times as slow, meaning that the root-mean-square speed of the hydrogen molecules are 4 times as fast.
12	A	By First law of thermodynamics, besides cooling, the internal energy can also be decreased by